SCHOOL OF COMPUTING (DEMO)

CSC2103 DATA STRUCTURES

SAMPLE FINAL EXAMINATION (written for this demo)

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Time allowed: 2 hours	Total marks: 100
Number of questions: FIVE (5)	Weighting: 50% of the final grade

INSTRUCTIONS TO CANDIDATES

1. Answer ALL questions.
2. Write your answers in the answer booklet provided.
3. Code may be written in JavaScript or clear pseudocode.
4. The marks for each part are shown in square brackets.

Question 1 (15 marks): Algorithm analysis and arrays

(a) Using the formal definition of Big O notation, show that $f(n) = 4n^2 + 10n + 7$ is $O(n^2)$. State suitable values for the constants c and n_0. **[5 marks]**

(b) State the time complexity of the code fragment below in Big O notation. Justify your answer. **[4 marks]**

```
for (let i = 1; i < n; i = i * 2)
  for (let j = 0; j < n; j++)
    count++;
```

(c) A dynamic array doubles its capacity whenever it is full. Explain why appending n elements to an initially empty dynamic array costs $O(n)$ in total, and state the amortised cost of a single append. **[6 marks]**

Question 2 (30 marks): Hashing

(a) Define the load factor α of a hash table. Explain why a hash table is resized when α exceeds a threshold, and describe what happens during a resize. **[4 marks]**

(b) The keys 18, 41, 22, 44, 59, 32, 31, 73 are inserted, in this order, into an initially empty hash table of size $m = 13$ using the hash function $h(k) = k \bmod 13$ and **linear probing**. Show the final contents of the table and state the number of probes needed to insert 31 and to insert 73. **[10 marks]**

(c) Repeat part (b) using **separate chaining** instead of linear probing. Draw the resulting table. **[6 marks]**

(d) Explain what primary clustering is and why it occurs with linear probing. Describe how double hashing reduces it. **[5 marks]**

(e) With linear probing, the expected number of probes for an unsuccessful search is approximately $\frac{1}{2}(1 + 1/(1 - \alpha)^2)$. Calculate this value for $\alpha = 0.5$ and $\alpha = 0.9$, and use your answers to recommend a maximum load factor. **[5 marks]**

Question 3 (30 marks): Heaps and priority queues

(a) State the two properties that together define a binary min-heap. **[4 marks]**

(b) A binary heap is stored in an array starting at index 0. Give the index of the parent, the left child and the right child of the node at index i. **[3 marks]**

(c) Insert the keys 35, 33, 42, 10, 14, 19, 27, 44, 26, 31, in this order, into an initially empty min-heap. Show the array after each insertion. **[8 marks]**

(d) Perform one extract-min operation on the final heap from part (c). Show every swap and the final array. **[6 marks]**

(e) Explain why building a heap bottom-up from n unordered keys takes O(n) time, while inserting the n keys one at a time takes O(n log n) time in the worst case. **[5 marks]**

(f) Name TWO applications of priority queues and explain briefly why a heap suits each one. **[4 marks]**

Question 4 (10 marks): Linked lists

(a) Compare a singly linked list with a dynamic array in terms of (i) access by index, (ii) insertion at the front and (iii) memory use. **[6 marks]**

(b) Write a function `reverse(head)` that reverses a singly linked list in place and returns the new head. State its time complexity. **[4 marks]**

Question 5 (15 marks): Binary search trees

(a) Insert the keys 45, 25, 65, 15, 35, 55, 75, 30, in this order, into an initially empty binary search tree. Draw the final tree. **[4 marks]**

(b) Give the in-order, pre-order and post-order traversals of the tree from part (a). **[6 marks]**

(c) Delete the key 25 from the tree in part (a), using the in-order successor. Draw the resulting tree. **[3 marks]**

(d) State the worst-case height of a binary search tree with n nodes and give an insertion order that produces it. **[2 marks]**

END OF QUESTION PAPER